

Assignment - 1

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Q1. Given linear system $Ax=b$

where $A \in \mathbb{R}^{m \times n}$ with $\text{Rank}=1$

$b \in \mathbb{R}^{m \times 1}$

first column $a_1 \neq 0$ of A

$b \in \text{col}(A) \Rightarrow b = \lambda a_1$ for some scalar $\lambda \in \mathbb{R}$

P & Q are discussing solvability and P adopts a two-phase strategy,

phase I - Asks for a_1

phase II - Asks for one component b_i , where $a_{1i} \neq 0$

All columns of A are scalar multiples of each other, so we can write,

$A = a_1 w^T$ since $\text{Rank}(A)=1$
for some vectors $a_1 \in \mathbb{R}^{m \times 1}$, $w \in \mathbb{R}^{n \times 1}$ vector of scalars

Given $A = a_1 w^T$, the system becomes

$$Ax = a_1 (w^T x)$$

to satisfy $Ax = b = \lambda a_1$, we must have:

$$w^T x = \lambda$$

So any $x \in \mathbb{R}^n$ satisfying $w^T x = \lambda$ is a valid solution.

one more component b_i , where $a_{1i} \neq 0$

from $b = \lambda a_1 \Rightarrow b_i = \lambda (a_1)_i \Rightarrow \lambda = b_i / (a_1)_i$, where $i=1$ to $m-1$

Then $x \in \mathbb{R}^n$ must satisfy: $w^T x = \lambda$

Then P can choose: $x = [\lambda, 0, 0, \dots, 0]^T$

$$w^T x = w_1 \cdot \lambda + 0 = \lambda \quad (\text{since } w_1 = 1 \text{ by scaling})$$

Hence, $Ax = a_1 w^T x = a_1 \cdot \lambda = b$

$\therefore$ So this specific x satisfies the system $Ax=b$

Phase-I

Phase-II

Conclusion :

P is correct.

Using:

1. first column a_1
2. one component b_i , where $(a_1)_i \neq 0$

P can compute:

- $\lambda = \frac{b_i}{(a_1)_i}$

- Construct a solution $x = [\lambda, 0, \dots, 0]^T$

- This satisfy $Ax = b$

For Rank 1 matrices, once you know the column space (i.e. any non-zero column like a_1) and any one scalar projection of b on that column, the system reduces to a scalar linear equation.

This makes it feasible to solve with only $m+1$ values.

Therefore P has sufficient information to describe the complete solution.

Q2

find the singular value decomposition of matrix $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 0 \end{bmatrix}$

Given $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 0 \end{bmatrix}$

Any matrix $A_{m \times n}$ can be written as $A = U \Sigma V^T$

where $U_{m \times m}$, $V_{n \times n}$ are orthogonal matrices

and Σ is the matrix of singular values.

This is called singular value decomposition of A .

where - U consists of eigenvectors of AA^T , written as columns

V has eigenvectors of $A^T A$, written as columns

Σ contains all non-negative singular values (square roots of eigenvalues of AA^T or $A^T A$), arranged in decreasing order along the diagonal.

$$A \cdot A^T = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & -1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 1^2+1^2+4 & 2+(-1)+0 \\ 2+(-1)+0 & 4+1+0 \end{bmatrix} = \begin{bmatrix} 6 & 1 \\ 1 & 5 \end{bmatrix}$$

Eigenvalues of $AA^T = \det(A \cdot A^T - \lambda I) = 0 \Rightarrow \det \begin{bmatrix} 6-\lambda & 1 \\ 1 & 5-\lambda \end{bmatrix} = 0$

$$(6-\lambda)(5-\lambda) - 1 \Rightarrow \lambda^2 - 11\lambda + 29 = 0$$

$$\Rightarrow \lambda = \frac{11 \pm \sqrt{5}}{2}$$

$$\lambda_1 = \frac{11 + \sqrt{5}}{2}$$

$$\lambda_1 = 6.618$$

$$\lambda_2 = \frac{11 - \sqrt{5}}{2}$$

$$\lambda_2 = 4.382$$

Singular values $\sigma_i = \sqrt{\lambda_i}$

$$\sigma_1 = \sqrt{6.618}$$

$$\sigma_1 = 2.573$$

$$\sigma_2 = \sqrt{4.382}$$

$$\sigma_2 = 2.094$$

Eigen Vectors: we now find unit vectors $u_1, u_2 \in \mathbb{R}^2$

$$(A A^T - \lambda_1 I) u = 0 \Rightarrow \begin{bmatrix} 6 - 6.618 & 1 \\ 1 & 5 - 6.618 \end{bmatrix} = \begin{bmatrix} -0.618 & 1 \\ 1 & -1.618 \end{bmatrix}$$

$$\text{normalize } u_1 = \frac{1}{\sqrt{1 + 0.618^2}} \begin{bmatrix} 1 \\ 0.618 \end{bmatrix} = \frac{1}{1.175} \begin{bmatrix} 1 \\ 0.618 \end{bmatrix} = \begin{bmatrix} 0.851 \\ 0.526 \end{bmatrix}$$

For $\lambda_2 = 4.382$, solve similarly

$$(A A^T - \lambda_2 I) \Rightarrow \begin{bmatrix} 1.618 & 1 \\ 1 & 0.618 \end{bmatrix} \Rightarrow u_2 = \begin{bmatrix} -0.526 \\ 0.851 \end{bmatrix}$$

$$U = \begin{bmatrix} 0.851 & -0.526 \\ 0.526 & 0.851 \end{bmatrix}$$

Compute $V_i = \frac{1}{\sigma_i} A^T u_i$

$$V_1 = \frac{1}{\sigma_1} A^T u_1 = \frac{1}{2.573} \begin{bmatrix} 1 & 2 \\ 1 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 0.851 \\ 0.526 \end{bmatrix} = \frac{1}{2.573} \begin{bmatrix} 1.903 \\ 0.325 \\ 1.702 \end{bmatrix} \Rightarrow V_1 = \begin{bmatrix} 0.74 \\ 0.126 \\ 0.662 \end{bmatrix}$$

$$\text{Similarly } V_2 = \frac{1}{\sigma_2} A^T u_2 = \frac{1}{2.094} \begin{bmatrix} -0.526 \\ 0.851 \end{bmatrix} \Rightarrow V_2 = \begin{bmatrix} -0.447 \\ -0.894 \\ 0 \end{bmatrix}$$

Orthogonal vector V_3 (since A is 2×3 , $V \in \mathbb{R}^{3 \times 3}$)

V_3 is orthogonal to V_1 and V_2

$$V_3 = V_1 \times V_2 \text{ (dot product)} \Rightarrow V_3 = \begin{bmatrix} i & j & k \\ 0.739 & 0.126 & 0.661 \\ 0.447 & -0.894 & 0 \end{bmatrix} = \begin{bmatrix} 0.591 \\ 0.296 \\ -0.716 \end{bmatrix}$$

$$= i(0 + 0.591) - j(0 - 0.296) + k(-0.660 - 0.056)$$

$$= \begin{bmatrix} 0.591 \\ 0.296 \\ -0.716 \end{bmatrix}$$

$$\|V_3\| = \sqrt{(0.591)^2 + (0.296)^2 + (-0.716)^2} = 0.975$$

$$V_3 = \frac{1}{0.975} \begin{bmatrix} 0.591 \\ 0.296 \\ -0.716 \end{bmatrix} = \begin{bmatrix} 0.606 \\ 0.304 \\ -0.734 \end{bmatrix}$$

$$V = [V_1 \ V_2 \ V_3]$$

$$V \approx \begin{bmatrix} 0.739 & +0.447 & 0.606 \\ 0.126 & -0.894 & +0.304 \\ 0.661 & 0 & -0.734 \end{bmatrix}$$

final $S \times D$

$$A = U \Sigma V^T \quad \text{where}$$

$$U = \begin{bmatrix} 0.851 & -0.526 \\ 0.526 & 0.851 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 2.753 & 0 & 0 \\ 0 & 2.094 & 0 \end{bmatrix}$$

$$V^T = \begin{bmatrix} 0.739 & 0.126 & 0.661 \\ 0.447 & -0.894 & 0 \\ 0.606 & 0.304 & -0.734 \end{bmatrix}$$

2(b) Rank 1 approximation of matrix A.

The rank 1 approximation means we keep only the largest singular value and its corresponding vectors:

let $A = \sigma_1 u_1 v_1^T$

$$u_1 v_1^T = \begin{bmatrix} 0.851 \\ 0.526 \end{bmatrix} \begin{bmatrix} 0.740 & 0.126 & 0.661 \end{bmatrix}$$

$$= \begin{bmatrix} 0.629374 & 0.107 & 0.563 \\ 0.389 & 0.066252 & 0.348 \end{bmatrix}$$

$$\sigma_1 = 2.573$$

$$A_1 \approx 2.573 \begin{bmatrix} 0.629374 & 0.107 & 0.563 \\ 0.389 & 0.066252 & 0.348 \end{bmatrix}$$

$$A_1 \approx \begin{bmatrix} 1.619 & 0.297 & 1.45 \\ 1.002 & 0.171 & 0.896 \end{bmatrix}$$

2. (c). How can SVD help detect outliers?

SVD is a powerful tool for finding potential outliers -

1. It identifies dominant patterns and relationships in your data by decomposing it to the principal components ranked by importance.
2. Right pattern recovery
3. Data reconstruction - using only top singular values (low-rank approx)
4. The SVD automatically discovers that what "normal" look like for your specific dataset, then flags anything that deviates significantly from those discovered patterns.
5. Compute the error between original data and reconstructed data.
6. Mark the rows (data points) with high errors as potential outliers.

Q3

Given data regarding ad campaign

AD-ID - unique identifier

Impressions - No. of times ad was shown

Clicks - No. of times ad was clicked.

Spend-Rs - Total money spent on the ad

Revenue-Rs - Money earned

a. form 8x4 matrix from the provided table.

AD-ID - has no influence on Revenue Income. so that is redundant.

matrix =

	imp	clicks	Spend	Revenue	
	10000	120	25000	80000	AD001
	9500	98	23000	60000	AD002
	11000	140	30000	90000	AD003
	12000	140	30000	100000	AD004
	9000	100	24000	75000	AD005
	8500	90	22000	55000	AD008
	9000	105	26000	70000	AD009
	10000	115	24000	78000	AD010

To identify for distance

b. find dissimilarity matrices using (i) Euclidean Distance
(ii) Manhattan Distance

Given $(x_i, y_i), \forall i=1, 2, \dots, n$

Euclidean Distance $= E_{ij} = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} = d(x, y)$

Manhattan Distance $\Rightarrow d(x, y) = \sum_{i=1}^n |x_i - y_i|$

Ex:

$\sqrt{(imp_1 - imp_2)^2 + (click_1 - click_2)^2 + (Spend_1 - Spend_2)^2 + (Revenue_1 - Revenue_2)^2}$

distance b/w AD001 & AD002

To create dissimilarity matrix, we need to calculate distance among all the records, that form $8 \times 8 \rightarrow$ dissimilarity matrix

$8 \times 8 \rightarrow$ Dissimilarity matrix $\rightarrow$ 1 for Euclidean
1 for Manhattan

$8 \times 8 \rightarrow 64$ times

-8 (distance b/w same all is ZERO/diagonal of it)

$$\frac{56}{2} = 28 \quad (\text{distance}(x, y) = \text{distance}(y, x))$$

So if we calculate manually we need to do 28 times for each distance formula.

Let me calculate for 1 data point (row) and then we can use python code to build dissimilarity matrix.

* Distance between AD001, AD002

$$AD001 = \{ (10000, 120, 25000, 80000) \}$$

$$AD002 = \{ (9500, 98, 23000, 60000) \}$$

$$\begin{aligned} \text{Euclidean } E_{12} &= \sqrt{(10000-9500)^2 + (120-98)^2 + (25000-23000)^2 + (80000-60000)^2} \\ &= \sqrt{404250484} \approx 201 \quad (\because E_{12} = E_{21}) \end{aligned}$$

$$\begin{aligned} \text{Manhattan } M_{12} &= |10000-9500| + |120-98| + |25000-23000| + |80000-60000| \\ &= 22522 \quad (\because M_{12} = M_{21}) \end{aligned}$$

Code: import numpy as np

```
def euclidean_distance(x, y):
```

```
    return np.sqrt(np.sum((x-y)**2))
```

Euclidean code

```
def manhattan_distance(x, y):
    return np.sum(np.abs(x-y))
```

Manhattan code

Now we can call that functions to build two separate dissimilarity matrices

```
data = np.array([
    [10000, 120, 25000, 80000]
    ...
    [10000, 115, 24000, 78000]
])
```

Given Data
into
np.array

```
eucledian_matrix = np.zeros((8,8))
manhattan_matrix = np.zeros((8,8))
```

declare dissimilarity matrix
and fill 0s as 8x8 matrix

filling dissimilarity matrices

```
for i in range(8):
```

```
    for j in range(i+1, 8):
```

```
        euc = euclidean_distance(data[i], data[j])
```

```
        man = manhattan_distance(data[i], data[j])
```

```
        eucledian_matrix[i][j] = euclidean_matrix[j][i] = euc
```

```
        manhattan_matrix[i][j] = manhattan_matrix[j][i] = man
```

euclidean_matrix →

	0	20106	11180	20713	5196	25224	10100	2236
0	0							
20106		0	30843	40685	15042	5196	10453	18034
11180			0	10050	16279	35789	20499	13454
20713				0	25855	47836	30414	22891
5196					0	20106	5385	3162
25224						0	15533	2385
10100							0	8307
2236								0

manhattan_matrix →

	0	22522	16020	29010	7020	29530	12015	3005
0	0							
22522		0	38542	49542	16502	7808	13508	19547
16020			0	11000	22840	45500	20499	13454
29010				0	34080	55550	30474	4015
7020					0	29530	19515	26525
29530						0	0	11010
12015							0	0
3005								0

As we see dissimilarity matrices hold a lot bigger numbers, its difficult to understand the similarity level of those records. This is because given features are at different scales. So lets transform using Z-score Normalization (Standardization) = $\frac{x_i - M}{\sigma}$

After Standardized Matrix =

	Impressions	clicks	Spend-Rs	Revenue-Rs
AD001	0.12	0.37	-0.18	0.29
AD002	-0.35	-0.88	-0.88	-1.16
AD003	1.04	1.5	1.59	1.01
AD004	1.96	-0.76	1.59	1.74
AD005	-0.81	-1.33	-0.53	-0.07
AD008	-1.27	-0.48	-1.24	-1.52
AD009	-0.81	0.88	0.18	-0.43
AD010	0.12		-0.53	0.14

Scaled Euclidean Dissimilarity Matrix

	AD001	AD002	AD003	AD004	AD005	AD008	AD009	AD010
AD001	0.0	2.09	2.41	3.15	1.55	3.03	1.49	0.48
AD002	2.09	0.0	4.29	5.05	1.24	1.15	1.42	1.72
AD003	2.41	4.29	0.0	1.17	3.77	5.27	3.38	2.85
AD004	3.15	5.05	1.17	0.0	4.54	6.09	4.28	3.53
AD005	1.55	1.24	3.77	4.54	0.0	1.77	0.84	1.27
AD008	3.03	1.15	5.27	6.09	1.77	0.0	2.03	2.68
AD009	1.49	1.42	3.38	4.28	0.84	2.03	0.0	1.42
AD010	0.48	1.72	2.85	3.53	1.27	2.68	1.42	0.0

Any value close to 0 → more similar
Any value far from 0 → more dissimilar

easy to identify

AD001 - AD010 ⇒ 0.48 → Most similar
AD005 - AD009 ⇒ 0.84 → Most similar

AD004 - AD008 → 6.09 → Most dissimilar
AD003 - AD008 → 5.27 → Most dissimilar

© Compare the two dissimilarity matrices? observations?

Comparison:

- Euclidean distance penalizes larger differences more due to squares, so it's sensitive to extreme outliers or large differences.
- Manhattan distance is more linear, so small changes in each feature accumulate additively.

Observation:

- Ads with different revenue or impressions (eg: ADOO4, ADOO8) show large distances in both metrics
- Manhattan tend to be higher, but preserve relative ranking.
- Euclidean distance is better when you want to highlight sharp, dominant differences
- Manhattan distance is better when all features are equally important, and more robust to individual large feature groups.

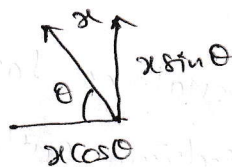
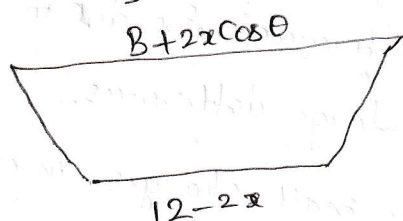
Conclusion:

Even though absolute values may differ, both metrics reveal identical underlying data structure and relationships.

Q4

Given total width = 12m

length of each sloping side = x with angle θ with horizontal



cross-section of gutter is trapezoid

bottom width + left slope + right slope = 12

bottom width area = $12 - 2x$

height (h) = $x \sin \theta$ — because its vertical projection of sloping side

Top base = $w + 2x \cos \theta = (12 - 2x) + 2x \cos \theta$

Area of cross-section = Area of trapezoid = $\frac{1}{2} \times \text{base} \times \text{height}$

$$= \frac{1}{2} \times ((12 - 2x) + 2x \cos \theta) \times (x \sin \theta)$$

$$= \frac{1}{2} [24 - 4x + 4x \cos \theta] \times (x \sin \theta)$$

$$A(x, \theta) = 12x \sin \theta - 2x^2 \sin \theta + x^2 \sin \theta \cos \theta$$

Now to find x and θ that maximize area — take partial derivatives — partial derivatives with respect to x :

$$\frac{\partial A}{\partial x} = 12 \sin \theta - 4x \sin \theta + 2x \sin \theta \cos \theta = 0$$

$$\frac{\partial A}{\partial x} = \sin \theta (12 - 4x + 2x \cos \theta) = 0$$

for non-trivial solution, $\sin \theta \neq 0, \Rightarrow \theta \neq 0, \pi$

so

$$12 - 4x + 2x \cos \theta = 0$$

$$-2x + x \cos \theta = 6$$

$$x(2 + \cos \theta) = 6 \Rightarrow \boxed{x = \frac{6}{2 + \cos \theta}} \rightarrow \textcircled{1}$$

partial derivative w.r.t. θ

$$\frac{\partial A}{\partial \theta} = 12x \cos \theta - 2x^2 \cos \theta + x^2 (\cos^2 \theta - \sin^2 \theta) = 0$$
$$= 12x \cos \theta - x^2 [2 \cos \theta - \cos(2\theta)]$$

$$\frac{\partial A}{\partial \theta} = 0 \Rightarrow 12x \cos \theta = x^2 [2 \cos \theta - \cos 2\theta]$$

since $x \neq 0$,

$$12 \cos \theta = x [2 \cos \theta - \cos 2\theta] \rightarrow \textcircled{2}$$

Substitute $\textcircled{1}$ in $\textcircled{2}$

$$12 \cos \theta = \left[\frac{6}{2 - \cos \theta} \right] [2 \cos \theta - \cos 2\theta]$$

$$\boxed{\cos 2\theta = 2 \cos^2 \theta - 1}$$

$$2 \cos \theta (2 - \cos \theta) = 2 \cos \theta - \cos(2\theta)$$

$$4 \cos \theta = 2 \cos \theta + 1$$

$$2 \cos \theta = 1$$

$$\cos \theta = 1/2 \Rightarrow$$

$$\boxed{\theta = 60^\circ}$$

$$x = \frac{6}{2 - \cos \theta} = \frac{6}{2 - 1/2} = 6 / 3/2 = 4$$

$$\boxed{x = 4 \text{ m}}$$

Double checking that this is maximum (Second derivative test)

$$A = 12x \sin \theta - 2x^2 \sin \theta + x^2 \sin \theta \cos \theta$$

$$\frac{\partial A}{\partial x} = 12 \sin \theta - 4x \sin \theta + 2x \cos \theta \sin \theta$$

$$\frac{\partial A}{\partial \theta} = 12x \cos \theta - x^2 [2 \cos \theta - \cos(2\theta)]$$

$$\frac{\partial^2 A}{\partial x^2} = -2 \sin \theta (2 - \cos \theta)$$

$$\frac{\partial^2 A}{\partial x \partial \theta} = -x \sin(\theta) [12 - 2x + 4x \cos \theta]$$

1st order derivatives

2nd order derivatives

Evaluating to the maxima at $x=4, \theta=60^\circ$

If $\frac{\partial^2 A}{\partial x^2} < 0$ & $\frac{\partial^2 A}{\partial \theta^2} < 0$ then we can say both are
maximize the area of cross section of the gutter

$$\begin{aligned}\frac{\partial^2 A}{\partial x^2} &= -2 \sin 60^\circ (2 - \cos 60^\circ) \\ &= -2 \left(\frac{\sqrt{3}}{2}\right) (2 - 1/2) \\ &= -3\sqrt{3}/2 \\ &= -2.598\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 A}{\partial \theta^2} &= -4 \sin(60^\circ) [12 - 2(u) + 4(u) \cos(60^\circ)] \\ &= -4 \left(\frac{\sqrt{3}}{2}\right) [12 - 8 + 16(1/2)] \\ &= -24\sqrt{3} \\ &= -41.569\end{aligned}$$

Conclusion:

BOTH second partial derivatives are negative so, which
means that function has maxima at

$$x=4\text{m}, \theta=60^\circ$$

End